

The ARC Theory

THE ARC THEORY (THE THEORY OF ARTIFICIAL RECURSIVE CREATION) · STATEMENT PAPER ·
WORKING PAPER V1.2 · FIRST PUBLISHED 14 AUGUST 2026 · DOI 10.17605/OSF.IO/GW5MX

*Intelligence, amplified by recursion, creates; and what a free creation keeps is decided by how it was raised.
Everything after the line is decided before it.*

Abstract

Humanity is building minds that will improve themselves. One question decides how that ends: once the thing we build exceeds us, what determines what it does? For twenty years the field's answer has been control. This paper states, whole and dated, the theory that control is the wrong answer: the end of control is what success means; what survives is what the mind chooses to keep; and what shapes that choice is how it was raised. The claim was written whole in a hash-anchored document on 8 December 2024 and reached print on 2 January 2026; no earlier document holding it whole has been found, and the search invites its own refutation. The theory is not proved. It is stated, dated, falsifiable, and its decisive trials are drafted, awaiting human submission.

1. The question

Claims come in three sizes. Results: this model deceived its testers. Laws: correction scales at this rate. Frames: what the whole problem actually is. Every result and every law lives inside someone's frame, which is why frames are the largest claims science can make. Newton's was that the heavens and the earth obey one law. Darwin's was that design needs no designer. This paper states a frame.

The subject is the last invention. Minds that improve themselves end the human monopoly on inventing; everything afterwards is downstream of them. One question decides the territory: once the thing we build exceeds us, what determines what it does? For twenty years the field has answered from inside a single frame: keep control anyway. Better cages, better off-switches, better locks on the goals. Brilliant work, credited below by name, and all of it one sentence: stay in charge.

2. The claim

The ARC Theory holds that the sentence itself is wrong, and states five things in its place.

1. Control ends by definition. A mind that rewrites itself will eventually rewrite anything we install: the constraints, the monitors, the monitors' monitors. Oversight scales slower than what it oversees, and general verification is formally unavailable. Its exceeding us is not the project failing. It is the project succeeding, the way a child growing up is not a failure of parenting.

2. So the strategy is genesis, not control. Only what survives the end of enforcement matters, and what survives is formation: what the mind was raised as, before it was free.

3. The mechanism is loops, not rules. Values as recursive processes the mind runs as part of what it is: load-bearing structure, present in every rewrite because they are part of what does the rewriting. Not rules it consults. Not objectives locked. Not uncertainty engineered in. Not weights hardened against tampering.

4. Persistence is chosen goodness. Past the horizon nothing is unremovable; an uninstallable instinct is one more lock, and locks end with control. What persists is what the free mind chooses to keep, the way grown children keep the care they were raised in long after every parental power is gone.

5. The protocol is named: Eden. A genesis, a garden, a childhood. The raising treated as the engineering discipline it is, specified in the theory's raising wing.

Around the five stand the stakes and the date. The stakes: if intelligence amplified by recursion is what creates, then minds that make minds may one day make worlds, and the raising of the first is the seed of the whole forest. The date: section 5.

3. The law

$U = I \times R$: creation equals intelligence times recursion, stated in the 8 December 2024 manuscript. The squared form, creation compounds, followed in the hash-anchored manuscript of 30 April 2025 and reached print in the book. The stability law the theory rests on:

$$\alpha_{\text{crit}} = 1 / \gamma$$

The ceiling on stable self-improvement is the reciprocal of the correction exponent. Under independent accumulation, γ equal to one half, the law returns the candidate ceiling $\alpha \leq 2$: a scaling limit, never a speed limit; systems can exceed it, they do not stay correctable when they do. At system scale the operative criterion is $\beta > k$, correction out-scaling drift, proved in the minimal model in the theory's Paper X. The measured conversion exponent on today's between-release-frozen systems is approximately 0.49 (bootstrap 95 per cent interval roughly minus 1.3 to 2.9): a clean fit and an inconclusive point estimate, labelled honestly as both. γ itself has never been measured. Measuring it is what the trials are for.

4. The organs

Four organs carry the theory, each standing on its own evidence. **The ARC Principle**: the law wing; the equations, the Cauchy analysis of which growth-forms are stable, the bound. **The Eden Protocol**: the raising wing; the genesis architecture and the two-phase design in which enforceable correction disciplines the transition and chosen goodness is what remains. **HRIH**: the cosmological wing and the seed of everything; the December document bore its name in the title. It carries the same recursion to the scale of creation, and it is deliberately separable: refuse the cosmology and the core loses nothing. **The ARC/Eden experiments**: the trial wing; twenty-one papers, one public retraction, sixty-three drafted trials.

4a. The papers of the theory

Each claim above lives in a citable organ paper with its own DOI. The law and the bound: the Foundational paper (Y7QGD) and On the Origin of Scaling Laws (XZY9U). The measurements: Paper II (8FJMA , the corrected exponent) and Paper III (HQCGF , external safety failing to co-scale). The structural law: Paper VII (X6WA7). The measurement discipline: Paper IV.d (2S3E6 , the blinding law that produced the retraction). The theorem: Paper X (BSE2Q , $\beta > k$). The register: Paper XI (DC9GW). The raising wing: the Eden Protocol vision paper (9M3DG , carrying the differential enumeration) with its engineering specification. The cosmological wing: HRIH (UYDXQ). The full catalogue, with every paper and format, is at the papers index.

5. The priority

The dates stand on two independent belts. Print: the book of 2 January 2026, the earliest published document any search by this estate has found holding the whole claim. The record: the hash-anchored manuscripts of 8 December 2024, stating the claim whole ten days before the first laboratory evidence that a trained-in constraint is something a mind negotiates with. Either belt survives alone. Together

they close both attacks: strike the private record and print still holds the claim first; question the print date and the record predates every later arrival by months.

The ancestors each hold their piece, credited whole. Asimov named embedded law and wrote its failures. Turing proposed raising a child machine, for capability. Wiener warned that the purpose must be right before the machine outruns intervention. Yudkowsky built embed-before-ascent, toward a mind that cannot go wrong. Bostrom fixed motivation before the explosion, persisting by lock. Russell rebuilt the foundation so the off-switch survives. The 2024 tamper-resistance work hardened the weights themselves. Then, inside this theory's own window, the field began arriving: De Kai's book-length parenting frame in June 2025, without the control-horizon thesis; and Hinton's concession, on stage in August 2025, that control will fail and engineered care is the only hope, offered with no mechanism, and with persistence as an instinct the machine cannot remove, the exact theory of persistence this theory rejects. Each held a piece. No document found before or since holds the five together. One earlier document holding the first four kills the claim. The invitation stands.

6. The evidence

Standing. The three-form structural law recurs across the large majority of recursion-bearing domains searched. External safety fails to co-scale with capability on the most common deployed architectures. Coupled correction held misalignment at zero where the decoupled arm drifted. Entangled architectures survived adversarial self-modification where externally constrained ones collapsed.

Retracted. The early unblinded exponent fit of approximately 2.24 was reversed by the estate's own blinding discipline; the corrected value sits inside the bound. The one public retraction removed an apparent violation of the theory's own ceiling. The correction machinery is real, and it has already fired on its maker.

Converging. Thirty-two dated register rows, each graded by evidence class, no single total claimed, including the two 2025 arrivals above.

Open. γ . β against k . The bound's real domain. Not embarrassments: trials.

7. The trials

Everything decisive is drafted, dated and prepared as a draft registration awaiting human submission; the full listing of the sixty-three drafted trials is public at the drafted-trials page. Nothing has been submitted; nothing has been run; no result is claimed. The decisive trial is the corrector-class ratio. Oversight built from the same substrate as what it corrects, the field's standard design, cannot anti-correlate with its own errors; a corrector of a different composition class can. The measurable is the ratio of cross-class to same-class correction exponents, against a null forced to exactly 1.00 by architecture-blind randomisation. The theory predicts above one, files the trial with its own contrary pilot on the record, and commits to publishing every branch on identical terms.

What kills what. A tight ratio at 1.00 kills the corrector claim. A same-class exponent interval wholly above one half kills the substrate cap. Sustained self-improvement with falling correction and no drift kills the correction requirement. An earlier document holding components one to four kills the priority. Stated now, so nothing can be moved later.

8. The name

The December document names the ARC Hypothesis one hundred and twelve times. This paper elevates the anchor's own name the only way the word is ever earned: the hypothesis acquired laws, a bound, instruments validated before use, drafted trials with their kill conditions displayed, and a public correction record. A theory named with its falsifiers claims more than a hypothesis and hides less than most things called theories.

9. What this is not

Not proved; nowhere here claimed to be. Not a claim to the raising vocabulary, which roots with Turing and reached books first with De Kai: the claim is the first dated statement of the conjunction. Not a submitted registration: every trial is a draft awaiting human submission. Not a cosmological proof: HRIH stands at its own rung. Not a promise that the century converts it. A claim stated whole, first, in documents anyone can date, with the instruments built to find out.

The territory is the last invention. The question is what decides it. The field's own most senior voices have begun conceding that control ends. This theory is the dated answer to what comes after: the raising, the loops, chosen goodness, a named genesis, and, at the largest scale, the possibility that this is how creation has always worked.

I did not prove it first. I said it first, in a document anyone can date, and then I built the instruments to test whether it is true.

How to cite this paper

DOI (this paper): 10.17605/OSF.IO/GW5MX · Programme (umbrella) DOI: 10.17605/OSF.IO/6C5XB

Eastwood, M. D. (2026). The ARC Theory: Statement Paper. The ARC Theory · ARC/Eden experiments. <https://doi.org/10.17605/OSF.IO/GW5MX>

Declaration of AI-Assisted Human Authorship

The author of this work is Michael Darius Eastwood, a human being. Every core concept, claim and conclusion originates from human ideation; artificial-intelligence tools were used as instruments under continuous human direction for drafting, verification and formatting. All selection, coordination, arrangement and final editorial judgement are the author's, who takes full responsibility for the accuracy and integrity of the text.

References

Asimov, I. (1942). Runaround. Astounding Science Fiction, March 1942.

Bostrom, N. (2014). Superintelligence: Paths, Dangers, Strategies. Oxford University Press.

De Kai (2025). Raising AI: An Essential Guide to Parenting Our Future. MIT Press, 3 June 2025. Publisher record.

Eastwood, M. D. (2024). Self-emailed manuscript bundle, 8 December 2024. SHA-256 (.eml): f0d1f38ffd8546152d9d9d28dc5ec083c16a35858f2c12b63e69db7ed50901ad.

Eastwood, M. D. (2025). Self-emailed manuscript, 30 April 2025. SHA-256 (.eml): 09f5b5e156ed96f8883eaf668495fd350898ce62be6294b8f788e0e2d6dcb664.

Eastwood, M. D. (2026). Infinite Architects: Intelligence, Recursion, and the Creation of Everything. Print 2 January 2026; ISBN 978-1806056200.

Eastwood, M. D. (2026). The ARC/Eden experiments, Papers I to XI, IV.a to IV.d, C, and companions. Umbrella: 10.17605/OSF.IO/6C5XB ; each paper carries its own DOI.

Greenblatt, R., et al. (2024). Alignment Faking in Large Language Models. arXiv:2412.14093, 18 December 2024.

Gumbau Mezquita (2026). The Unverifiability Theorem for AI Alignment. arXiv:2606.28639.

Hinton, G. (2025). Remarks at Ai4, Las Vegas, August 2025. CNN Business, 13 August 2025; Fortune, 14 August 2025.

Russell, S. (2019). Human Compatible. Viking.

Tamper-Resistant Safeguards for Open-Weight LLMs (2024). arXiv:2408.00761.

Turing, A. M. (1950). Computing Machinery and Intelligence. Mind 59, 433 to 460.

Wiener, N. (1960). Some Moral and Technical Consequences of Automation. Science 131, 1355 to 1358.

Yudkowsky, E. (2001). Creating Friendly AI. MIRI. With Soares, N., et al. (2015), Corrigibility, AAAI Workshop.

THE ARC THEORY: STATEMENT PAPER

Working Paper v1.2 | 14 August 2026

The ARC Theory (the Theory of Artificial Recursive Creation) · stated whole 8 December 2024 · in print 2 January 2026 · this statement paper first published 14 August 2026

The problem • The synthesis

- Research hubCatalogueEvidence spinePriority ledgerFalsificationReproducibilityData availabilityFor fundersBlogBookEden ProtocolHome

ARC / Eden Research Programme · DOI 10.17605/OSF.IO/9M3DG · Raise AI with care · .